

ddPCR™ Assay Design Service Assays

Catalog #	Description
12008226	ddPCR Assay Design Service FAM Assay, 1,000 x 20 µl reactions
12008227	ddPCR Assay Design Service HEX Assay, 1,000 x 20 µl reactions
12008228	ddPCR Assay Design Service Cy5 Assay, 1,000 x 20 µl reactions
12008229	ddPCR Assay Design Service Cy5.5 Assay, 1,000 x 20 µl reactions
12008230	ddPCR Assay Design Service 450 µM Primer, 50 µl
12008231	ddPCR Assay Design Service 125 µM FAM Probe, 50 µl
12008232	ddPCR Assay Design Service 125 µM HEX Probe, 50 µl
12008233	ddPCR Assay Design Service 125 µM Cy5 Probe, 50 µl
12008234	ddPCR Assay Design Service 125 µM Cy5.5 Probe, 50 µl
12008235	ddPCR Assay Design Service 375 µM Dark Probe with Iowa Black FQ, 50 µl
12008236	ddPCR Assay Design Service 375 µM Dark Probe with Iowa Black RQ-Sp, 50 µl
12020293	ddPCR Assay Design Service 125 µM ROX Probe, 50 µl
12020307	ddPCR Assay Design Service 125 µM ATTO 590 Probe, 50 µl

For research purposes only.

Description

ddPCR Assay Design Service (ADS) Assays are designed specifically for Droplet Digital™ PCR (ddPCR) based on customer requests. ddPCR ADS Assays can encompass a wide variety of assays (for example, translocation, mutation detection, multiplex mutation screening, etc). An assay design may require a combination of multiple catalog numbers and unique assay IDs. Information required to formulate an assay for a particular assay design request can be obtained from the assay information pages available online at [bio-rad.com/digital-assays](https://www.bio-rad.com/digital-assays). These assays should be validated prior to use.

Ordering Information

ddPCR Assays can be ordered online at [bio-rad.com/digital-assays](https://www.bio-rad.com/digital-assays). Required ddPCR Supermixes can also be ordered at [bio-rad.com](https://www.bio-rad.com).

Storage and Stability

ddPCR Assays are stable for 12 months when stored at 4°C protected from light. The 20x assay mix can be kept at –20°C for long-term storage.

Kit Contents

The ddPCR ADS Assay is a concentrated primer, probe, or primer-probe mix. Refer to Instructions for Use section.

Equipment and Reagents

- QX100 or QX200™ Droplet Generator (catalog #1864002) or Automated Droplet Generator (#1864101)
- QX100, QX200 (#1864003), or QX600™ (#12013328) Droplet Reader
- QX ONE™ Droplet Digital PCR System (#12006536)
- C1000 Touch Thermal Cycler with 96–Deep Well Reaction Module (#1851197) or PTC Tempo Deepwell Thermal Cycler (#12015392)
- PX1 PCR Plate Sealer (#1814000)

- For 1-2 targets per well in FAM and/or HEX assays, ddPCR Supermix for Probes (No dUTP) (#1863023, 1863024, or 1863025) is recommended
- For 1-2 targets per well in Cy5, Cy5.5, ROX, and/or ATTO 590 assays or >2 targets per well, ddPCR Multiplex Supermix (#12005909, 12005910, or 12005911) is recommended

Go to [bio-rad.com](https://www.bio-rad.com) for additional supermixes for other applications.

Refer to the QX200 Droplet Generator Instruction Manual (10031907), Automated Droplet Generator Instruction Manual (10043138), or QX ONE Droplet Digital PCR System and QX ONE Software User Guide (10000116512) for ordering information on consumables such as oils, cartridges, gaskets, plates, and seals.

Instructions for Use

ddPCR ADS Assays #12008226, 12008227, 12008228, and 12008229 are 20x concentrated, ready-to-use primer-probe mixes (9 µM primers and 5 µM probe).

Adding 4 µl of the required number of assay components from #12008230, 12008231, 12008232, 12008233, 12008234, 12008235, 2008236, 12020293, or 12020307, and water to a final volume of 100 µl results in a concentration of 18 µM primer, 5 µM probe, and 15 µM dark probe. A further 20x dilution in the ddPCR reaction results in 900 nM primer, 250 nM probe, and 750 nM probe. Additional guidelines can be found in the product inserts for any ddPCR Supermix.

For more information about assay design and optimization for Droplet Digital PCR, refer to the Droplet Digital PCR Applications Guide (bulletin 6407) and the Rare Mutation Detection Best Practices Guide (bulletin 6628).

Data Acquisition and Analysis

Follow instructions for data acquisition and analysis based on the droplet reader in use.

- For the QX200 Droplet Reader, refer to the QX200 Droplet Reader and QX Manager Software Standard Edition User Guide (10000107223) or the QX200 Droplet Reader and QX Manager Software Premium Edition User Guide (10000163733)
- For the QX600 Droplet Reader, refer to the QX600 Droplet Reader and QX Manager Software Standard Edition User Guide (10000153877) or the QX600 Droplet Reader and QX Manager Software Premium Edition User Guide (10000153878)
- For the QX ONE ddPCR System, refer to the QX ONE Droplet Digital PCR System and QX ONE Software User Guide (10000116512) and the QX ONE Software User Guide for Standard Edition (10000116655) or Regulatory Edition (10000116656)



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